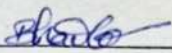


SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	T. Bhanu Teja	Reg. No.:	192472259
Section / Batch:	CSE - AI	Date:	19-08-26

Code of Conduct

I T. Bhanu Teja (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

Annexure A – Case Study

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences. The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} | \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution $P(\text{is})=0.66$ and $P(\text{drives})=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
 - economic (120 occurrences)
 - growth (450 occurrences)
 - increases (210 occurrences)
 - employment (380 occurrences)

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

- Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

C03-AT2

DS10301 -
Natural Language
ProcessingCase Study 1: Smart mobile keyboard prediction system.1. MLE of Bigram probability ($P(\text{science}|\text{data})$)

$$P(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$$

$$C(\text{data}) = 3, \quad C(\text{data science}) = 3$$

$$\therefore P(\text{science}|\text{data}) = \frac{C(\text{data science})}{C(\text{data})} = \frac{3}{3} = 1.0$$

Interpretation

The probability is 1.0 or 100%, meaning that in the given corpus, every occurrence of data is followed by science.

2. Back off model for "data science improves"

The sequence is

data $\rightarrow$ science $\rightarrow$ improves

The bigram science improves is unseen in training corpus.

Order: Trigram $\rightarrow$ Bigram $\rightarrow$ Unigram

Since science improves is unseen, the model backs off to

 $P(\text{improves})$

The word improves does not occur in given corpus.

$$\therefore C(\text{improves}) = 0$$

$$\therefore P(\text{improves}) = 0$$

So, with the unsmoothed statistics provided in question.

$$P(\text{improves} | \text{science}) = 0$$

and consequently,

$$P(\text{data science improves}) = 0$$

3. Deleted Interpolation for "data science is"
 → Deleted interpolation combines trigram, bigram, and unigram probabilities.

$$P_{\text{interp}} = \lambda_1 P_{\text{tri}} + \lambda_2 P_{\text{bi}} + \lambda_3 P_{\text{uni}}$$

Given:

$$\lambda_1 = 0.5, \lambda_2 = 0.3, \lambda_3 = 0.2$$

Trigram probability: $P(\text{is} | \text{data, science}) = \frac{C(\text{data science is})}{C(\text{data science})}$

Since: $C(\text{data science is}) = 2$ & $C(\text{data science}) = 3$

$$P_{\text{tri}} = \frac{2}{3} = 0.6667$$

Bigram probability

$$P(\text{is} | \text{science}) = \frac{C(\text{science is})}{C(\text{science})} = \frac{2}{3} = 0.6667$$

Unigram probability

$$P(\text{is} | \text{science}) = \frac{2}{12} = 0.1667$$

$$P_{\text{interp}} = (0.5)(0.6667) + (0.3)(0.6667) + (0.2)(0.1667) \\ = 0.3335 + 0.2000 + 0.03334$$

$$P_{\text{interp}} \approx 0.5667$$

4. Entropy of the Distribution:

$$P(\text{is}) = 0.66, P(\text{driver}) = 0.33$$

Entropy is $H(x) = -\sum P(x) \log_2 P(x)$

$$\therefore H = -[0.66 \log_2 (0.66) + 0.33 \log_2 (0.33)]$$

$$\therefore H \approx 0.923 \text{ bits}$$

Interpretation:

→ Low entropy $\Rightarrow$ high prediction confidence

→ High entropy $\Rightarrow$ greater uncertainty.

Case Study 2 :- AI - powered Customer Support Chatbot.

1. Penn tree bank - POS Tagging

(i) $\Rightarrow$ Book a flight ticket now.

<u>word</u>	<u>POS tag</u>	<u>Explanation</u>
Book	VB	verb
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

$\Rightarrow$ Book / VB a / DT flight / NN ticket / NN now / RB

(ii) $\Rightarrow$ This book is interesting

$\Rightarrow$ This / DT book / NN is / VBZ interesting / JJ

2. HMM Likelihood of "book" as VB

$$\Rightarrow P(\text{VB}, \text{book}) = P(\text{start} \rightarrow \text{VB}) \times P(\text{book} | \text{VB})$$

$$\text{Given: } P(\text{start} \rightarrow \text{VB}) = 0.5$$

$$P(\text{book} | \text{VB}) = 0.6$$

$$\text{Therefore: } P(\text{VB}, \text{book}) = 0.5 \times 0.6$$

$$P(\text{VB}, \text{book}) = 0.30$$

$$P(\text{VB}, \text{book}) = 0.5 \times 0.6$$

$$P(\text{VB}, \text{book}) = 0.30$$

for comparison: $P(\text{NN}, \text{book}) = 0.5 \times 0.4 = 0.20$

$$\text{Therefore } \Rightarrow 0.30 > 0.20$$

So the HMM favours:

book / VB.

Interpretation: The model gives a higher probability to VB because the emission probability of "book".

3. Rule - Based vs Stochastic - HMM Tagging

Rule - Based Tagging

Advantages:-

- * Simple to understand
- * Does not require large training data
- * Rules are easy to interpret.

Limitations:-

- * Requires many manually written rules.
- * Difficult to handle exceptions
- * Less flexible with ambiguous words.

HMM Tagging

It considers:-

- * word (tag) probabilities
- * Tag transition probabilities
- * context.

→ A hybrid approach combining rules and statistical model can provide even better performance.

4. Importance of standardized POS Tagsets:

→ Standardized POS tagsets such as Penn Treebank provide a consistent representation of grammatical information.

Role in chat bot Systems:-

The verb in a sentence can indicate the user's action. Ex: "Book a flight".

Context understanding:

POS tags help distinguish ambiguous words

Book a flight → VB

This Book → NN

Conclusion:- POS tagging improves intent detection, context understanding, response generation, and overall chatbot accuracy.

Case Study 3: News Analytics and POS Tag Correction System.

1. Apply the transformation rule:-

Economic /JJ growth /NN increases /NNS employment /NN

Here Rule is change NNS to VBZ if the preceding word is tagged as NN.

Updated POS Sequence:-

economic /JJ growth /NN increases /VBZ emp

Explanation: The sentence is

Economic growth increases employment.

→ Here, Economic growth is the subject, increases is the verb, and employment is the object.

2. Analysis of the Initial Tagging Error

Initial tag assigned:

increases /NNS

→ When NNS represents a plural noun.

However, economic growth increases employment.

Correct Structure

<u>Word</u>	<u>Tag</u>	<u>Grammatical Rule</u>
economic	JJ	Adjective
growth	NN	subject noun
increases	VBZ	verb
employment	NN	object noun

The sentence follows:

Subject + verb + object

∴ The transformation rule correctly changes NNS → VBZ.

3. Word frequency distribution:

<u>word</u>	<u>frequency</u>
economic	120
growth	450
increases	210
employment	380

∴ Total = 1160

$$P(\text{word}) = \frac{C(\text{word})}{1160}$$

<u>Word</u>	<u>frequency</u>	<u>probability</u>	<u>Percentage</u>
economic	120	0.1034	10.34%
growth	450	0.3879	38.79%
increases	210	0.1810	18.10%
employment	380	0.3276	32.76%

4. Transformation - based Tagging and entropy.

Before transformation

economic / JJ growth / NN increases / NNS employment / NN

error is increases / NNS

Apply vb2

NNS → VB2

After transformation

economic / JJ growth / NN increases / VB2 emp

Role of entropy ; $H(X) = - \sum P(x) \log_2 P(x)$

Suppose before

Correction the system considers:

$$P(\text{NNS}) = 0.5, P(\text{VB2}) = 0.5$$

$$\text{Then } H = - [0.5 \log_2(0.5) + 0.5 \log_2(0.5)]$$

$$H = 1 \text{ bit.}$$

→ This indicates uncertainty between the two tags